

Nuts and Bolts of New Ventures: Session 3, Part 1: Legal Issues

Joseph Hadzima

Lecture notes organized by [LazyingArt LLC](#) with [Video2Book](#)

Original lecture by Joseph Hadzima for MIT OpenCourseWare. Lecture notes organized by [LazyingArt LLC](#).

Chapter 1

Legal Issues

These notes follow Joseph Hadzima's MIT OpenCourseWare lecture on legal issues in new ventures, with curation in this volume by LazyingArt LLC. The lecture is practical from the first minute. We are not being invited to become amateur lawyers. We are being taught to notice the places where a venture can lose ownership, lose leverage, or create a tax problem simply by doing the right thing too late.

Hadzima therefore organizes the hour as a sequence of timing problems. He begins with the life cycle of the venture, moves through intellectual property and ownership, tightens around patents as exclusion rights and strategic assets, and then closes with entity choice, tax elections, and the Section 83 trap. The unifying question is always the same: what has to be done early, before success itself makes the legal position more expensive?

1.1 Framing Legal Issues Along the Venture Life Cycle

Hadzima lowers the temperature before he raises it. This is background information, not legal advice. The reason is twofold and worth preserving exactly in spirit: legal outcomes are fact-dependent, and the law itself changes. The lecture is thus not a catalogue of eternal rules. It is a hazard map. If we see a particular pattern, we should know enough to stop, ask questions, and get help before stumbling into a pothole.

He also tells us, almost apologetically, that he is going to overload the room. That warning matters. It establishes the pedagogical contract of the lecture. He has one shot to put the structure in our heads; the slides and video can be revisited later. So we should expect density, but not randomness.

The first structural device is the venture life cycle. Across the top sit stages: idea, pre-outside-financing, first financing, second financing, and beyond. Down the left run recurring topic rows: intellectual property, legal entity, people, and financing. Even without the slide itself, the logic is clear. We are being asked to read legal issues in temporal order. What matters at the idea stage is not identical to what matters after a priced round.

That is why he begins where founders usually postpone thinking: "I have an idea," and "I am getting ready for outside financing." At that stage we are already forced to ask who owns the idea, what kind of protection exists, what legal body should hold the rights, and whether the company has been formed soon enough to avoid avoidable trouble later. The investor will ask those questions

in diligence. Hadzima wants the founders to ask them while the answers are still cheap.

1.2 Why Intellectual Property Matters, and What Kind of Protection We Actually Have

The lecture's first quantitative move is motivational. Why worry about intellectual property at all? Because the value of firms has moved away from plant and equipment and toward intangibles. Hadzima uses a 2015 slide in which the intangible share of S&P 500 market capitalization is about 84%. The mathematical skeleton of his hierarchy is

$$\text{Intellectual property} \subset \text{Intellectual capital} \subset \text{Intangible assets}, \quad \text{Intangible share of S\&P 500 market cap} \approx 84\% \quad (1.1)$$

This nested relation matters. Intangible assets include people, processes, ideas, and organizational capability. Intellectual capital is the narrower set of ideas and processes. Intellectual property is narrower still: it is the subset that the law lets us protect, license, assign, or lose.

Definition 1.1. Intellectual property is the legally protectable subset of a venture's intellectual capital. In Hadzima's practical register, it is the part of intangible value that can be fenced, transferred, and tested in financing.

He then walks through the ladder of protection in a very deliberate order. The lecture starts with "none," which is important. We are meant to remember that legal protection is not automatic. We may simply have ideas that we choose not to protect. From there he moves to trade secret, trademark or service mark, copyright, and patent. Each step is motivated by what it protects, what it does not protect, how long it lasts, and what it tends to cost.

A concise summary of the lecture's timing structure is

$$\text{Term}_{\text{trade secret}} = \infty \quad \text{while secrecy is maintained}, \quad (1.2)$$

$$\text{Term}_{\text{patent, U.S.}} = 20 \text{ years from filing}, \quad (1.3)$$

$$\text{Intent-to-use window} \leq 3 \text{ years}, \quad (1.4)$$

$$\text{Copyright duration} \approx 70 + \text{ years}, \quad (1.5)$$

$$\text{Copyright registration cost} \approx \$100 - \$500. \quad (1.6)$$

Trade secret is the simplest to state and one of the hardest to manage well. It lasts as long as we keep the information secret. That is why Hadzima immediately connects it to non-disclosure agreements. The trade secret is not magic; it is a secret that gives market advantage, and the law only helps if we behave as though secrecy matters. The lecture's practical rule is not to explain the secret sauce unless there is a specific reason, a specific recipient, and usually an agreement around that disclosure.

Trademark and service mark are different. They identify source, not technical function. A trademark is for goods, a service mark for services. Rights arise through use, and federal registration strengthens them. Hadzima keeps the founder on practical ground here. Before registration, we

mark the claimed brand with TM or SM; only after federal registration does the registered-mark symbol become available. The mark can be a word, symbol, phrase, sound, or even color. The lecture's red-sole-shoe example is meant to make that breadth memorable.

Two practical consequences follow. First, a mark should be fanciful rather than merely descriptive. "Apple" for computers is protectable in a way that "micro dose" for low-dose aspirin is not. Second, one should search before investing heavily in branding. Rights arise through use, but Hadzima emphasizes the *intent-to-use* filing precisely because founders may spend a great deal on design and launch preparation only to learn too late that the mark is unavailable.

He also preserves a subtlety that matters to founders who think federal registration erases everything that came before. It does not. The senior unregistered user can survive in the geographic area where use began earlier. His McDonald's example makes the point vividly: a small earlier user may survive locally, even after the national chain registers federally.

Copyright is introduced with equal care. It arises upon creation, not registration. If Hadzima writes a letter, he already has a copyright in the expression. But the right protects expression, not idea or function. That is why Shakespeare's treatment of a love triangle is protectable while the abstract idea of a love triangle is not. The distinction is not philosophical decoration. It explains why copyright is a natural fit for literature and music, and a more awkward fit for software.

The lecture then becomes founder-facing again. Registration is cheap relative to patents, but if one wants to sue, registration matters. The work must be deposited, and in software that raises obvious anxiety: must we deposit source code? Hadzima's answer is procedural rather than doctrinal. There are ways to deposit redacted portions so that authorship can be proved without giving away everything.

His Aero Maps stories serve the same didactic function. Phantom roads are inserted so copying becomes detectable. The Boston directory story is even sharper: the city stopped buying updated directories because it had copied the old one. That anecdote does not merely illustrate infringement. It demonstrates why a venture should put notices on what it wants to protect, and why one should not confuse "I made it" with "I can prove I own and can enforce it."

The most important copyright lesson for founders is ownership. The author of a work owns it unless the work falls within employment or a valid work-for-hire arrangement. Hadzima's import-export software story is painful for exactly that reason: a company can spend roughly half a million dollars building software and still not own it if the contract never assigned ownership. The companion lesson is symmetric. When a startup consults for a large company, the services agreement will often attempt to transfer everything created. Founders must negotiate the difference between the deliverable and the preexisting tools, modules, and know-how used to produce it.

The lecture adds two modern edges to this section. One is open-source compliance: if we modify open-source code in ways that trigger license obligations, that becomes part of diligence. The other is artificial intelligence. Hadzima's lecture-era caution is simple enough to keep: copyright requires human authorship, AI alone is not the author, and the boundary of "sufficient human authorship" remains unsettled.

1.2.1 Question & Answer

Why worry about intellectual property if we only have an idea?

Because at the earliest stage the venture is mostly intangible. The people, the process, the know-

how, the technical insight, and the first sketches of product-market fit are the company. If value has shifted toward intangibles, then ownership and protection are not side issues. They are the first structural questions. Before we ask how fast the venture can grow, we must ask what exactly it owns, what it can protect, and what later investors will be able to underwrite.

The lecture's rhythm also matters here. Hadzima does not begin with patent doctrine. He first makes the room care about intangible value. Only then does the taxonomy of protections make economic sense.

1.3 Patents as Exclusion Rights, Timing Constraints, and Strategic Assets

Once patent appears in the taxonomy, the lecture changes pace. The earlier categories are compared and illustrated; patent is analyzed. Hadzima calls it a limited-time monopoly, but immediately sharpens that phrase into its legal meaning. A patent is a federally granted exclusion right. It lets us prevent others from making, using, selling, or importing what falls within the claims. It does not, by itself, guarantee that we are free to practice every commercially useful embodiment of our invention.

Proposition 1.2. *A patent is a right to exclude, not a complete affirmative permission to practice an invention.*

Proof. Take Hadzima's cup example. Inventor *A* patents a vessel to hold a liquid. Inventor *B* later patents a handle. A commercial cup with a handle practices both inventions. Inventor *A* cannot commercialize the handled cup without infringing *B*'s handle patent. Inventor *B* cannot commercialize the handled cup without infringing *A*'s vessel patent. Each can exclude; neither has unilateral freedom to practice the combined product. A cross-license is required. $\square$

The real-estate analogy now becomes exact enough for the lecture's purposes. The claims are the fence. They define what is inside the property right. But a fence is not a road. It marks exclusion, not universal freedom of use.

We should also preserve the deeper social logic Hadzima mentions. Patent law is a trade: society grants a limited monopoly in exchange for disclosure. That is why patent "cuts both ways." It gives a powerful exclusion right, but only after we explain the invention well enough that others can learn from it. The system is justified not because monopolies are generally loved, but because the disclosure permits later cumulative invention.

The lecture's checklist for patentability is standard but should remain explicit: the invention must be new, non-obvious, useful, and within patentable subject matter. It must also survive the timing traps created by disclosure and sale. Hadzima's compact timing rule is

$$\text{U.S. patent filing window after an enabling disclosure or an offer for sale} = 1 \text{ year.} \quad (1.7)$$

The word "enabling" does real work. "I have anti-gravity boots" is not enabling; it tells us nothing about how the device works. A detailed public explanation may be enabling. Once the disclosure is enabling, or once the invention has been offered for sale, the one-year U.S. clock begins. In many other jurisdictions the rule is harsher: pre-filing disclosure may destroy foreign patent rights

immediately. Hadzima's story of the Egyptian team that published its result in a journal before considering patent consequences is included precisely to show how a scientific success can become a commercialization loss.

Non-obviousness receives a similar practical treatment. Hadzima does not turn it into doctrinal abstraction. He points instead to arguments of the form: the literature said this could not be done, and the market is now buying it, so it could not have been obvious to a person of ordinary skill in the art. The point is not that commercial success automatically proves non-obviousness. The point is that the analysis sits inside a real evidentiary world.

The lecture also marks a historical shift: first-inventor-to-file. The timing pressure is no longer merely "invent first." It is "file first, provided you are an inventor." That is why collaborations, joint ventures, and ambiguous middle-ground inventions become dangerous. Each side may think the jointly developed improvement is still in discussion while the other side is already racing to the office.

Before leaving the doctrinal core, Hadzima inserts another essential founder reminder: just as the author owns the copyright, the inventor owns the patent unless rights have been assigned. Companies therefore insist on invention disclosure and assignment agreements. The practical rule is simple: if the company needs to own the invention, the assignment must not be left to implication.

This leads directly to freedom to operate. A founder can spend time and money building a product only to discover that someone else's patent blocks launch. The freedom-to-operate opinion is the tool Hadzima mentions here. It is not magic protection and not a judicial decree of safety. It is a law firm's reasoned view that the proposed activity does not obviously infringe. That opinion also matters because willful infringement can produce treble damages, and so documented caution changes litigation posture.

1.3.1 Question & Answer

If we have a patent, are we free to practice the invention?

No. We are free only with respect to the claim boundary we own. Other patents may still cover indispensable components, adjacent improvements, manufacturing methods, or commercial embodiments. That is why Hadzima insists on keeping two questions distinct. "Can we get rights?" is a patentability question. "Can we launch without being blocked?" is a freedom-to-operate question.

This is one of the lecture's most important separations of mechanism from recommendation. The mechanism is exclusion by claim. The recommendation is to ask about infringement before market entry, not after.

1.4 Patent Process, Prior Art Mapping, and Cost Discipline for Startups

From here the lecture pivots from doctrine to operating strategy. If patents are exclusion rights in a crowded field, then the founder's real problem is no longer just "what can be patented?" but "what should be patented, when, at what cost, and in what competitive terrain?"

Patent mapping is introduced through an arithmetic failure. Hadzima describes sending out for an outside prior-art search. It took about six months. During those six months roughly 5000 new U.S.

patents were issuing per week. The rough consequence is

$$5000 \text{ new U.S. patents/week} \times 26 \text{ weeks} \approx 1.3 \times 10^5 \text{ new patents.} \quad (1.8)$$

The precise number is only a back-of-the-envelope estimate, but the strategic conclusion is solid: a static search can be obsolete by the time it returns.

That motivates the patent map. Hadzima describes it carefully. A box is a patent or patent application. The left edge of the box marks issue or publication date. A tail extending left marks filing date. The lines connecting boxes are citations. These citations matter in a patent-specific way. They are not merely academic references. Prior art must be disclosed to the patent office, and willful failure to disclose relevant prior art can threaten the validity of an otherwise strong patent because it may be treated as a fraud on the office.

Tesla is the lecture's main illustration. One can look at a single Tesla patent and see both the prior art behind it and the later patents that cite it. One can also look at the portfolio over time and infer where the company is extending a core invention. A cluster of later patents citing a core patent may tell us where the company is building a continuation or extension strategy.

The deeper lesson is business-facing. Hadzima gives two opposite stories from SBIR-style commercialization work. In one case a cluster of patents suggested a thicket to avoid; a sparser region suggested a better entry point. In another case the same kind of cluster looked attractive, because the founders believed they had solved the very problem that those clustered filings were circling. In both cases the map functioned not as a legal ornament but as a tool for choosing a beachhead market and even for deciding whether acquisition-by-portfolio might be the right strategy.

That is the point at which the lecture asks what to patent. Large firms may reward sheer filing activity. Startups cannot afford that luxury. The founder's question is not "what claims could possibly be obtained?" but "which filings add value, create leverage, support licensing, or protect a market worth entering?"

When should we file? Hadzima's answer is layered. We file before losing rights through public disclosure or sale. We file in time to win the first-inventor-to-file race. We file in time to support a licensing or acquisition conversation with actual issued or pending rights. And if full filing is too expensive too early, we use the provisional system as a disciplined placeholder.

$$\text{Provisional priority period} = 1 \text{ year,} \quad (1.9)$$

$$\text{Provisional filing fee}_{\text{micro}} \approx \$64. \quad (1.10)$$

The lecture is precise about what a provisional does and does not do. It establishes a filing date. It need not contain claims. But it must contain a meaningful description of the invention. Only what is described gets the earlier priority. This is why Hadzima's fruits-and-vegetables example matters. If protein becomes part of the invention later, and protein was not described in the provisional, the original filing date does not magically expand to cover the new matter.

That makes the provisional less like a complete patent and more like a disciplined temporal marker. Hadzima even mentions the startup version of this: a conference paper can be turned into a provisional by filing it with a cover sheet, provided it truly describes the invention well enough. For a cash-constrained startup, that can be the difference between preserving rights and losing them.

His own example from IP Vision makes the discipline concrete. Development ran in six-week cycles. Before deployment, the team asked what had been built, what had been thought through but not yet implemented, and what was important enough to file. Provisional applications became a routine timing tool rather than a rare heroic event.

The structure of a full application also enters here. Hadzima describes it almost as a term paper: field of invention, background, summary, detailed description, and then claims. The crucial doctrinal point returns: the detailed description must disclose the invention and its best mode of practice. The claims then define the boundary. Because claim drafting is the most technical part of the process, this is where he wants founders to spend money on qualified patent counsel rather than improvising.

Costs now become explicit:

$$\text{Patent preparation cost} \approx \$5,000 - \$15,000, \quad (1.11)$$

$$\text{Outlier patent-preparation bill} \approx \$35,000, \quad (1.12)$$

$$\text{Foreign patent cost}_{10 \text{ major countries}} \approx \$330,000 - \$500,000. \quad (1.13)$$

The \$35,000 anecdote is not a statistical estimate; it is a governance lesson. The co-founder kept calling the patent lawyer directly, adding changes, and running the meter. Hadzima's response was managerial, not theoretical: no one talks to the patent lawyer without coming through a single gatekeeper. Patent cost is partly doctrinal, partly procedural, and partly behavioral.

Prosecution cost is open-ended in the same way. The examiner often rejects first claims, and Hadzima even says that can be a good sign, because immediate allowance may mean the applicant tried to claim too little. But every back-and-forth round costs money. Fixed-fee patent packages may therefore yield claims that are formally allowed and strategically weak.

Foreign filing raises the economic question again. If an exclusive licensee can be brought in early, part of the bargain may be that the licensee carries later patent costs. This turns timing of licensing into a financing strategy for IP itself.

1.4.1 Question & Answer

Should we patent this idea, or disclose it to block others?

Sometimes the right move is not to patent everything, but to patent the core and disclose the periphery. Hadzima's "picket fence" example explains why. Suppose we hold a strong core patent on the vessel to hold a liquid. Others might patent the top, the sleeve, the handle, or the insulating shell and thereby block commercial use of the core invention. If the core patent is strong, a cheap counterstrategy is to disclose those peripheral possibilities without claiming them. The disclosure becomes prior art and makes those later blocking patents harder or impossible to obtain.

This is not a universal rule. It depends on confidence in the strength of the core patent. But it is exactly the kind of startup arithmetic the lecture wants us to learn: spend carefully, disclose strategically, and think in terms of market control rather than filing count.

Tesla is the larger version of the same point. Hadzima uses Tesla not to preach "open patents," but to broaden the notion of patent strategy. If the goal is to sell electric vehicles, complementary

infrastructure matters. Giving some patent access away may stimulate the ecosystem the company needs. The legal instrument is still patenting; the business objective is now ecosystem design.

1.5 Working With Lawyers and Moving University IP Into a Startup

At this point the lecture compresses the IP discussion into founder checklists. Employees should sign invention disclosure and assignment agreements. Consultants should be governed by proper work-for-hire or assignment language. Non-disclosure agreements should be used where appropriate, but with eyes open: venture investors often refuse them because they see many similar companies and do not want later claims that they misused one founder's disclosure when funding another.

The operational principle remains the same. Do not disclose the secret sauce until the last responsible moment. Make sure the company owns what it thinks it owns. Understand the patent landscape over time, not once. Preserve rights by filing on time.

Hadzima uses even the provisional-application discussion to make a cultural point. Large companies sometimes resist provisional discipline because it is hard to track. Startups cannot afford that luxury. Their problem is not internal paperwork elegance. Their problem is losing rights while moving too slowly.

Then comes the lecture's comic but serious interlude on lawyers. Legal expertise is bought by the hour. If we dump a lab notebook on the desk and say "write me a patent," we are buying expensive confusion. If we wait until the one-year deadline is nearly gone and demand a filing by Monday, we are buying emergency pricing. If instead we prepare invention disclosures, read patents in the area, draft the technical description, and let counsel focus on claims and legal framing, we turn legal spend into leverage.

The next founder question then arrives naturally: what happens when the IP comes from a university? Hadzima had been asked about this on a founders' panel, and he answers by inserting a quick institutional history. Before Bayh-Dole, federally funded university inventions belonged to the federal government, were managed from Washington, and were typically licensed non-exclusively. For a startup that wants to build a business around a specific technology, this was a poor commercialization regime.

$$\text{Pre-Bayh-Dole commercially licensed patents} \approx 5\%. \quad (1.14)$$

Bayh-Dole changes the geometry. Universities retain ownership, file patents, license them, prefer small businesses, give the government a royalty-free license, build commercialization programs, and share royalties with inventors. The modern technology licensing office is the institutional consequence of that change.

MIT appears in the lecture as a comparatively founder-friendly case. Hadzima describes the Technology Licensing Office as easier to work with than many counterparts because its first priority is getting inventions into the world, not maximizing immediate revenue. That difference in mission shapes terms and process.

The ownership rules matter. Significant use of MIT facilities or MIT-administered funds typically means MIT owns the invention. MIT does not assign the IP outright; it licenses. Sponsors may have first licensing rights. The practical moral is plain: if we are building around university technology, we must know whether the invention belongs to the company, the founder, the sponsor, or the

university. Investors will eventually force the issue if the founders do not.

Hadzima's Ali example and his Kendall Square explanation show how this becomes operational. If the work was not done with significant use of MIT facilities or funds, the founder may obtain a letter clarifying non-ownership by MIT. If the invention does belong to MIT, the startup typically develops it across the street, in commercial space, not in university labs. That physical move mirrors the legal move.

1.5.1 Question & Answer

How do we take university IP into a startup without quietly mis-owning it?

By bringing it through the front door. Hadzima's recommended device is often an option to obtain a license. The option gives the startup time to explore markets, talk to investors, and raise capital without pretending that the technology has already been transferred into the company. It is the institutional analogue of the provisional patent application: a way to preserve position while uncertainty is still being resolved.

The lecture preserves several useful lecture-era financial ranges:

$$\text{University license issue fee} \approx \$50\text{K} - \$150\text{K}, \quad (1.15)$$

$$\text{University running royalty} \approx 3\% - 5\%, \quad (1.16)$$

$$\text{MIT startup equity stake} \approx \text{single-digit } \%. \quad (1.17)$$

He also describes a specifically startup-oriented equity practice in which the university may take a small equity position and keep that percentage fixed through the first outside financing round. In his example, if MIT holds 5% and an investor later buys 20%, MIT still sits at 5% through that first round rather than being diluted immediately with the founders. That is not a universal law of licensing. It is a lecture-era practice point, and it changes how founders should think about early capitalization.

1.6 Entity Choice, Tax Elections, and the Section 83 Trap

The lecture now tightens again. After asking who owns the technology, Hadzima asks what legal body owns the company. If founders do nothing, they may already have a general partnership: one or more persons carrying on a business for profit. That requires no written agreement, which is precisely why it is dangerous. Liability is joint and several. Ownership, absent a contrary agreement, follows money rather than founder effort. Neither feature is hospitable to a venture-backed startup.

The practical recommendation is therefore immediate incorporation, usually as a corporation, and often in Delaware if repeated outside financing is likely. Delaware matters here not as brand mystique but as a jurisdiction with well-developed corporate law and comparatively predictable outcomes. Hadzima also insists on corporate formalities. A corporation only protects if it is actually treated as a separate legal person in records, signatures, and operations.

Tax then enters through that corporate personhood. A corporation taxed under Subchapter C bears tax at the corporate level, and distributions to shareholders are taxed again. A Subchapter S election changes the structure by making the income pass through to the owners.

$$\text{C-corp asset sale : corporate-level tax + shareholder-level tax,} \quad (1.18)$$

$$\text{S-corp asset sale : pass-through to the owners without corporate-level tax.} \quad (1.19)$$

The eligibility conditions are narrow enough that Hadzima states them explicitly:

$$\text{S-corp shareholder cap} < 100, \quad \text{classes of stock} = 1, \quad \text{S-election filing window} \approx 3 \text{ months after incorporation} \quad (1.20)$$

A venture capital investor usually destroys S status because the shareholder rules are restrictive and only one class of stock is permitted. But Hadzima does not let the audience dismiss the election as irrelevant. Founders say: we will be loss-making for years, so why worry now? His answer is the exit. Small acquisitions are often structured as asset sales. In that case, the C corporation pays tax on asset gain, while the S corporation does not bear corporate-level tax in the same way.

The lecture gives two vivid numbers here: an illustrative exit-tax difference on the order of 13%, and the DIVA story in which an early S election reportedly saved about \$2,000,000. These are example numbers, not timeless constants, but they preserve the lecture's logic: early entity and tax choices can matter only at the end, which is exactly why founders neglect them.

Then Hadzima marks the climax audibly: if we are falling asleep, wake up. This is the big trap. Section 83 says that if we receive property in connection with services, we recognize ordinary income equal to the fair market value of the property minus what we paid for it. In the lecture's notation bank, the core formula is

$$I_{\text{ordinary}} = \text{FMV} - P. \quad (1.21)$$

To make the tax consequence explicit, we may write

$$T = \tau(\text{FMV} - P), \quad (1.22)$$

where T is the tax owed and τ is the marginal tax rate used for the example. Hadzima gestures at "37% or whatever" and then computes with 40%, so we keep the symbolic formula and treat 40% as a rough example rate.

Now the lecture turns the formula onto the founders themselves. Suppose an investor pays \$1,000,000 for 50% of the company. What is the founder's 50% worth? Hadzima first lets the room infer a simple answer: perhaps about \$500,000. If the founder paid essentially nothing for those shares, then

$$I_{\text{ordinary}} \approx \$500,000, \quad (1.23)$$

$$T = 0.40 \times \$500,000 = \$200,000. \quad (1.24)$$

He then pushes the room toward an even harsher inference. The wording is damaged in the transcript, but the intended arithmetic is clear enough: if the founder's block is instead treated as worth about \$1,000,000, then

$$I_{\text{ordinary}} \approx \$1,000,000, \quad (1.25)$$

$$T = 0.40 \times \$1,000,000 = \$400,000. \quad (1.26)$$

The economic violence of the example is the point. Founder stock can create a personal tax bill long before the founder has cash liquidity.

Then comes the second blow: vesting. If the stock vests over time, the income may be measured when the stock vests rather than when it is first issued. So the tax base grows with success. The lecture's timing structure is

$$T(t_v) = \tau(\text{FMV}(t_v) - P), \quad \text{FMV}(t_v) \gg \text{FMV}(t_0), \quad (1.27)$$

where t_0 is the early issuance date and t_v the later vesting date. Hadzima's plain-English version is better than any abstraction: when the company begins, the idea is a dime a dozen; when the stock vests two years later and the company has done well, the shares may be worth \$100 each. That is exactly when the tax becomes lethal.

His California CFO story is included to show that this is not theoretical. A later executive asked for the 83(b) election paperwork, and the company suddenly realized it had issued a large amount of vesting stock without understanding the tax consequences. What looked like a successful company was also a ticking time bomb.

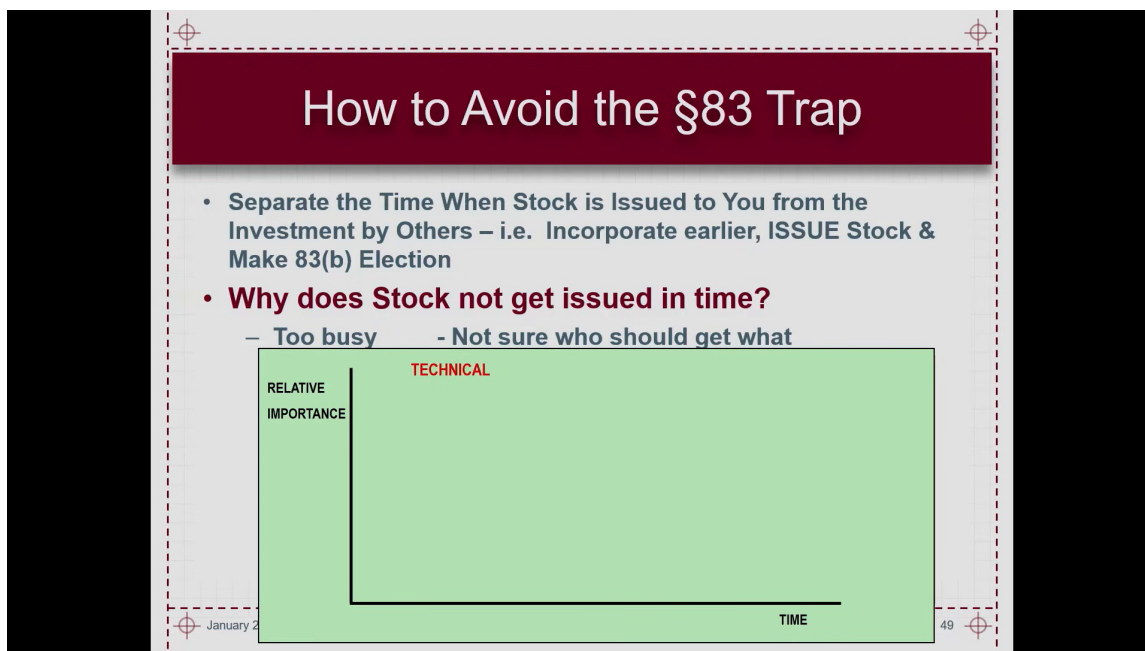


Figure 1.1: Section 83 trap slide with timing-versus-priority chart. The slide itself gives the operational recommendation: separate founder stock issuance from later outside investment and make the 83(b) election.

The screenshot matters because it preserves the lecturer's own wording. The top bullet says, in essence, separate in time the issuance of founder stock from the later investment by others:

incorporate earlier, issue stock, and make the 83(b) election. The second bullet asks why stock does not get issued in time. The visible answers are embarrassingly human: the founders are too busy, and they are not yet sure who should get what.

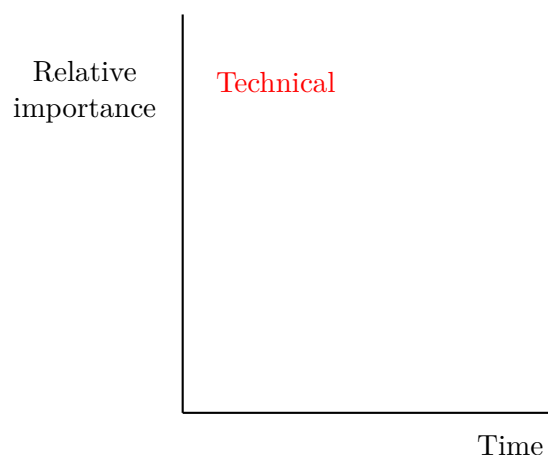


Figure 1.2: A pocket-scale reconstruction of the visible timing sketch. The frame supports only a qualitative claim: technical work is early and highly important, and that very urgency helps crowd out timely stock allocation.

1.6.1 Question & Answer

Why does stock not get issued in time, and how does an 83(b) election avoid the trap?

Because early ventures postpone the awkward split conversation while technical work, customer work, and financing consume attention. By the time the founders resolve who gets what, an outside investment or other valuation event may already have made the stock expensive for tax purposes. The 83(b) election changes the measuring time. It aims to tax the stock at the earlier, lower-value moment of transfer rather than the later, higher-value moment of vesting. But that only works if the stock has actually been issued, and if the issuance can be separated in time from the later event that makes valuation obvious.

That is why the lecture's recommendation is temporal, not merely formal: incorporate early, issue the stock early, let later cash investment occur later, and then file the 83(b) election promptly.

1.7 Founder Equity as Negotiated Structure, Not Myth

The lecture does not end by closing the tax folder and declaring victory. It spills, deliberately, into founder culture. How much equity should each founder have? Hadzima refuses to turn this into mythology. The amount of equity and who gets it is, in his words, about culture and negotiation.

His Keenan Systems example serves as a counter-myth. One founder kept all of the stock, paid employees very well, and later distributed bonuses when the company was acquired. That was a culture, not a universal fairness theorem. The lecture uses it to break the habit of thinking that there is a single morally canonical founder split.

The variables Hadzima wants on the table are concrete: value of past contribution, value of future

contribution over a specified time horizon, ownership of IP, sacrifice, commitment, reputation, opportunity cost, and the signaling effect of who is actually all-in. His professor-and-postdoc story preserves that method. Each person writes down what they contribute and what they give up. The professor brings name recognition, perhaps access to capital, perhaps one consulting day a week under MIT policy. The postdoc gives up a salary path and perhaps an academic career. The conversation becomes fairer not because someone solved it algebraically, but because the actual variables became visible.

The lecture then returns to numerical structure in the option-pool discussion. The equity compensation pool is not a decorative cap-table entry. It is a reserve sized for the next hiring horizon, typically two to three years and tied to the headcount plan. Hadzima's lecture-era range is

$$\text{Equity compensation pool after the first outside round} \approx 12\% - 18\%. \quad (1.28)$$

The lower end corresponds to a company that already has more of the needed senior team. The upper end corresponds to a company that still must recruit experience. Once again the lecture translates culture into timed structure. Who gets the pie and how large are the slices? We are supposed to have reasons, not slogans.

The ending remains intentionally unfinished. Hadzima notes that later lectures and slides will treat ownership examples, dilution, and financing instruments in more detail. That incompleteness is part of the chapter's rhythm. Founder equity is introduced here not as a closed calculus, but as the bridge from legal formation into financing.

1.8 Summary

The lecture moves from a high-level claim about intangible value to a founder-level tax trap, but the thread tying the whole argument together is timing. We ask when an intangible becomes property, when a disclosure becomes enabling, when a provisional must become a full application, when university ownership must be formalized by option or license, when a corporate election matters only because an exit may later occur, and when founder stock becomes taxable income.

If one lesson remains after the details blur, it is this: ventures are often injured not by exotic legal doctrine, but by ordinary legal rules encountered too late. Ownership, disclosure, assignment, filing, incorporation, election, vesting, and valuation all sit on a clock. Hadzima's practical achievement in this lecture is to make us see that clock before it starts billing us.